

Jaiveer Bassi

707-656-6227 | jaiveerbassi@yahoo.com | [linkedin.com/in/jaiveer-bassi](https://www.linkedin.com/in/jaiveer-bassi) | github.com/imjbassi | imjbassi.github.io

Experience

Physical Intelligence

May 2026 – Present

AI Robot Operator

San Francisco, CA

- Triage recurring hardware-software faults across the robotics stack, trace service logs to isolate root causes, and document reproducible findings for engineering teams.
- Evaluate policy rollouts against task criteria, identify failure modes and edge cases, annotate executions, and enforce data-quality thresholds used in model development.
- Generate high-quality demonstration data for general-purpose robotics foundation models by teleoperating robotic arms through manipulation, sorting, and assembly tasks.

Stanford University

Sep 2026 – Present

Stanford SSG Fellow

Stanford, CA

- Conduct mentor-guided independent research by developing a core curiosity question, evaluating primary scientific literature, and formulating testable hypotheses.

Handshake AI

Apr 2026 – Jun 2026

Contracted AI Response Evaluator

Remote

- Evaluated LLM-generated code and technical responses across Python, C#/.NET, AI/ML, and software engineering for correctness, instruction following, and technical clarity.

Castline Studio

Jan 2026 – Apr 2026

Operator & Developer – Resin 3D Printing, Next.js, TypeScript

Vallejo, CA

- Operated resin 3D-printing production for an e-commerce business generating \$2K+/month across 100+ product listings, including calibration, curing, post-processing, and tolerance control.
- Built an order-management platform with client-side STL parsing for automated geometry, volume, dimension, and print-cost calculations.

Kaiser Permanente Regional Laboratory

Oct 2023 – Jun 2024

Software Developer

Berkeley, CA

- Developed a full-stack application using React, Express, RESTful APIs, and SQL for uploading, processing, and reviewing clinical laboratory instrument data across multiple departments.
- Added validation and error logging to a 50+ column instrument-data workflow; diagnosed faulty recordings and documented recurring failure patterns to improve pipeline reliability.

Selected Projects

CAN Bus Fault-Injection & Diagnostics Rig | C++, Arduino, MCP2515 (SPI), SocketCAN

- Assembled a three-node CAN network of Arduino microcontrollers driving MCP2515 CAN controllers over SPI, with a USB-CAN interface and a custom wiring harness, then induced bus-off, arbitration, connector, wiring, and termination faults.
- Mapped physical failure modes to reproducible SocketCAN error states and multimeter evidence in a published diagnostic catalog.

Scribbot Desktop Drawing Arm | C++, Arduino Nano, PCA9685 (I2C), OpenSCAD

- Designed in OpenSCAD and built a 3D-printed four-servo drawing arm controlled by an Arduino Nano driving a PCA9685 PWM controller over I2C.

GELLO-Style Leader-Follower Teleoperation Rig | Arduino, ADC Sensing, Servo Control

- Built a four-DOF leader-follower mechanism for approximately \$30 by assembling linkages, wiring potentiometer-based joint encoders, and calibrating direct joint-space control without inverse kinematics.

π 0 on a Budget | OpenPI, LoRA, LeRobot, JAX, Arduino

- Built a synchronized data pipeline recording 30 FPS video and joint commands from a custom 4-DOF teleoperation arm, with conversion into a LeRobot-compatible dataset.
- Implemented an RTX 4070 workflow for π 0-FAST fine-tuning, memory profiling, policy serving, closed-loop evaluation, and a real-time dashboard of live Arduino telemetry and predicted action chunks.

UR5e Fault Injection & Diagnostics Harness | RTDE, URSim, Fault Injection

- Built a control and fault-characterization harness that drives simulated robot motion and logs synchronized joint telemetry while testing connection failures, stale handles, hangs, and native crashes.

Research Publications

When Does INT8 Actually Accelerate Host-CPU Inference?

- Audited MobileNetV2 INT8 conversion and packaging, showing that INT8 latency depends on backend delegation and quantization granularity; v1.2.0 archived on Zenodo.

One Byte, One Rank Reversal: Auditing HumanEval Pipeline Sensitivity

- In a pinned five-model RTX 4070 reproduction, StarCoder2-3B scored 3/164 against its published 31.7%; removing one trailing prompt newline raised it to 49/164 and restored the ranking.

Not All Bad Demonstrations Are Equally Bad

- Ran 660 controlled policy fits across five demonstration failure modes, two policy families, ten seeds, and transition-matched controls; under review at the CoRL 2026 LfC workshop.

Sampling Budget Biases Apparent Metastable-State Counts in Molecular Simulation

- Used controlled synthetic landscapes and two independent alanine-dipeptide trajectories to show how sampling budget can inflate apparent state counts; under review at JCTC.

Technical Skills

Languages: C++, Python, C#, TypeScript, JavaScript, SQL, Bash

Embedded & Hardware: Arduino, microcontrollers, SPI, I2C, CAN/SocketCAN, PWM servo and actuator control, sensor wiring, PCB rework, soldering, multimeter diagnostics

Robotics & Systems: RTDE, URSim, ROS2, real-time control, fault injection, telemetry analysis, Linux, Git, Docker, Kubernetes, PostgreSQL

Machine Learning: PyTorch, JAX, LeRobot, OpenPI, LoRA, ONNX, NumPy, model evaluation, CUDA

Education

Grand Canyon University

M.S. in Computer Software Engineering

- Capstone: Developed an AI model-packaging CLI that converts PyTorch, TensorFlow, ONNX, and scikit-learn models into deployable Docker images or Python packages.

University of Silicon Valley

B.S. in Computer Science

- Capstone: Developed a MetaHuman lip-sync and audio-animation pipeline in Unreal Engine for real-time playback.

Certifications

Accelerated Computing in Modern CUDA C++ (Sep 2026)

Cisco Network Support and Security (Sep 2025)

Stanford AI and Machine Learning in Healthcare (Jun 2025)

HackerRank Software Engineer Certificate (Mar 2026)

IBM Artificial Intelligence Fundamentals (Aug 2025)

Google IT Support Professional Certificate (May 2021)